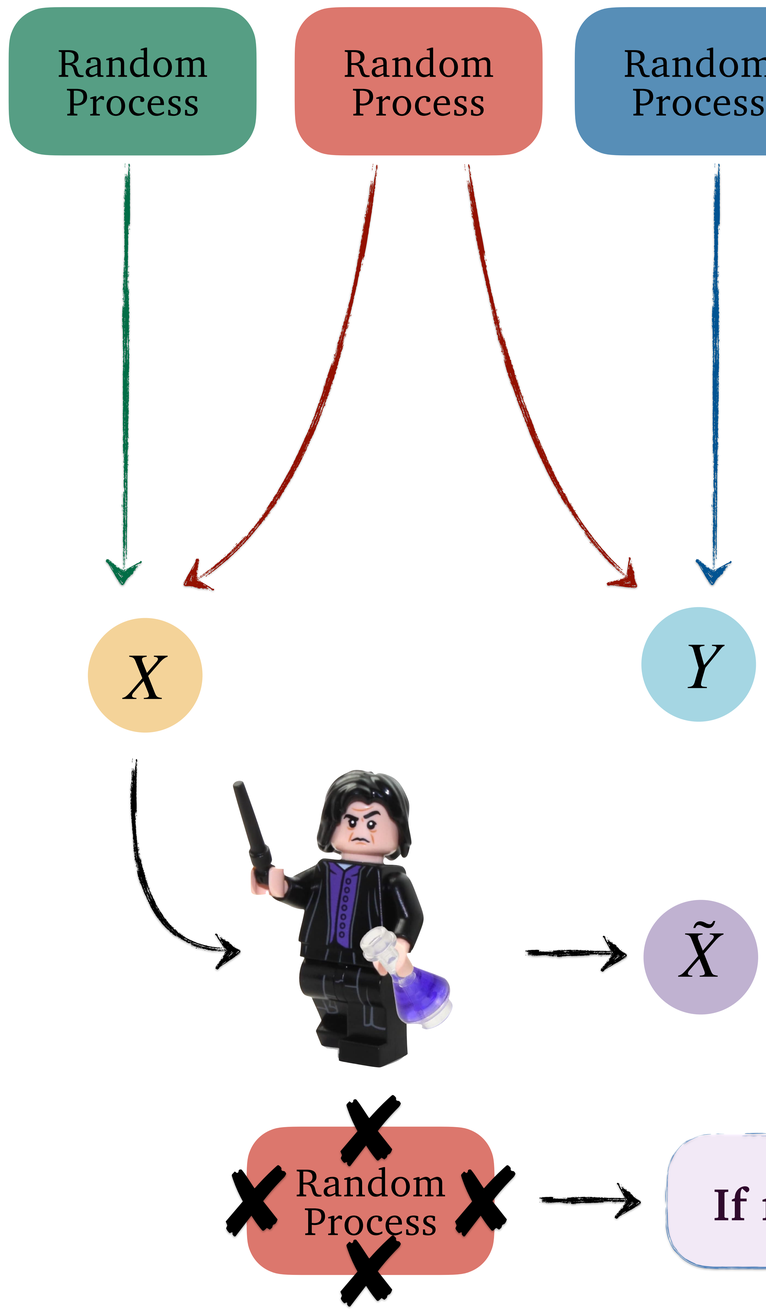
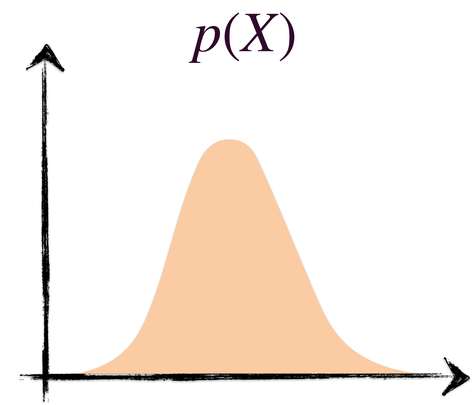
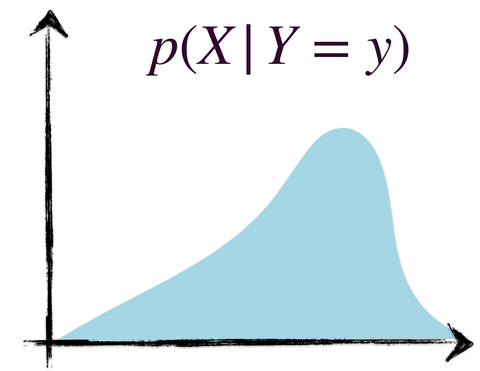
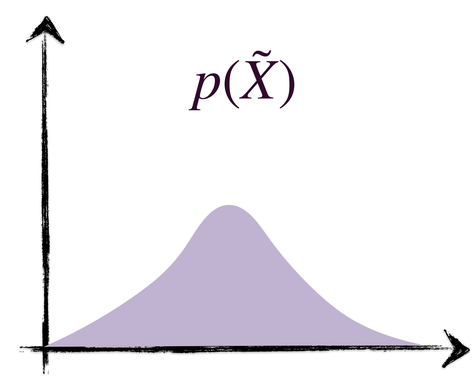
When Do We Expect Trade-offs After Erasure ?

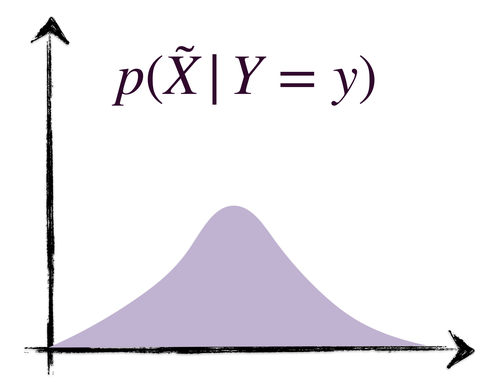




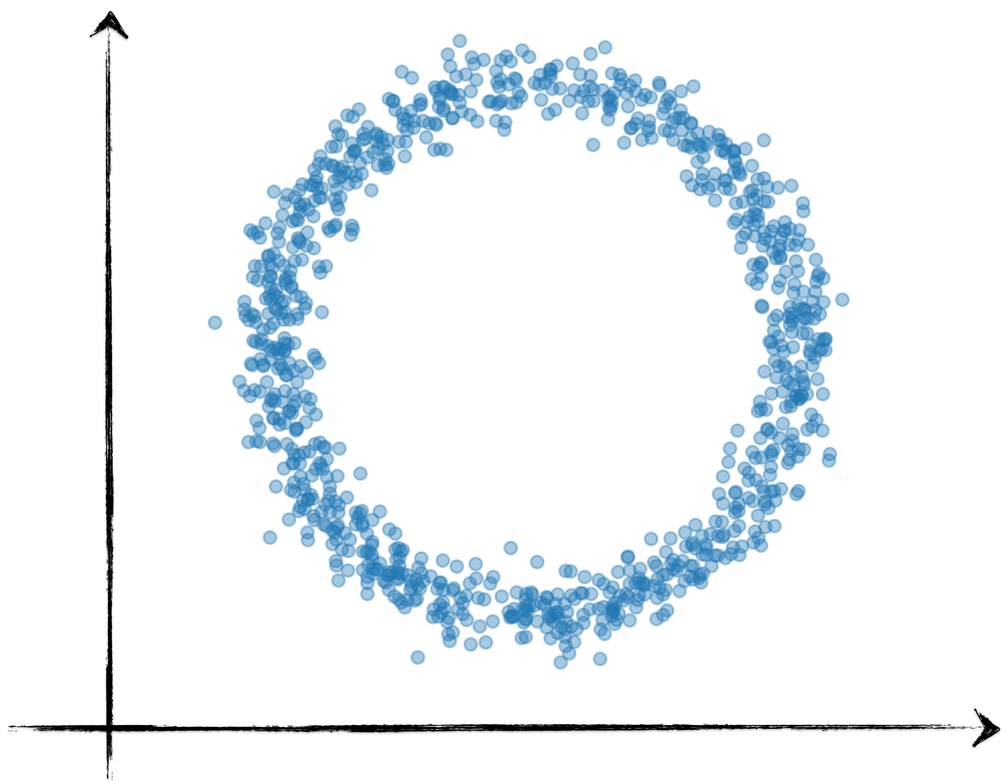


$p(X \mid Y) = P(X) \iff$ **Statisitcal Independence**





If required by down stream task → expect a trade-off

Why Linear Statistical Dependence Is Not Enough ?

$$\textbf{correlation} = \frac{\textbf{Cov}(X, S)}{\sigma_X \, \sigma_S} = \frac{\mathbb{E}\left[(X - \mu_X)(S - \mu_S)\right]}{\sigma_X \, \sigma_S}$$